

Time Series Analysis: Predicting Demand in US Power Markets

The electric grid provides reliable access to energy for homes and businesses across the world. Its stability is crucial for our modern life, enabling everything from healthcare to the economy. We are now faced with a situation of aging grid infrastructure that is not well-equipped for a renewable energy transition. In addition, several factors are driving an increase in demand for electricity including data centers, reshoring of manufacturing, and EV charging. Needless to say, there will be huge investment in new generation capacity, transmission, and grid technologies in the near future.

Traditionally, the electrical grid operated as a single vertically-integrated utility - encompassing generation, transmission and distribution. This model focused on delivering electricity from centralized generation plants outwards, and was not built for distributed renewable generation. A series of rule changes by the FERC in the late 90s allowed for this utility-monopoly model to change. It led to the creation of multiple Independent System Operators (ISO) and Regional Transmission Organizations (RTO) that facilitate wholesale power markets. See the attached map of US ISO/RTOs for reference.

The problems facing the electric grid are inherently technical and quantifiable. The emergence of power markets means that there is an abundance of information on important grid metrics used by all types of organizations: regulators, developers/operators, consumers, and traders. And, a lot of this data is collected at regular intervals creating an interesting application of time series analysis. The aim of this project is to allow you to begin playing around with electricity market data, predicting factors such as total grid demand.

To explore the project further, access the repository here:

<https://github.com/maxstclair/DS4002-CS3>